

Transshipment problem

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Solution of the practice quiz

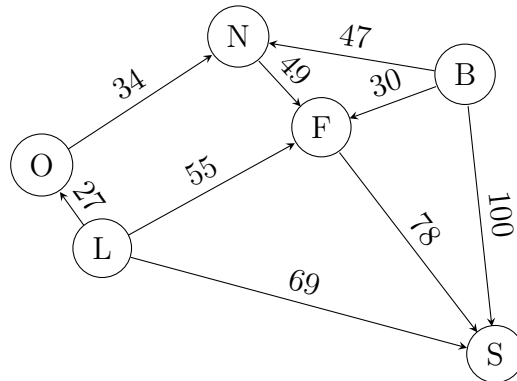
1. A transshipment problem has the form:

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \quad \forall i \in \mathcal{N},$$
$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A},$$

where $\mathcal{A}$ is the set of n arcs, $\mathcal{N}$ is the set of m nodes, $s \in \mathbb{R}^m$ represents the supply/demand of flow, $c \in \mathbb{R}^n$ represents the cost of transporting a unit of flow on each arc and $\ell \in \mathbb{R}^n$, $u \in \mathbb{R}^n$ represent the lower and upper capacity of each arc respectively. The decision variables of the problem are denoted x_{ij} , that is representing the flow on arc (i, j) .



The objective function is as follows:

$$\min_{x \in \mathbb{R}^9} 34x_{ON} + 27x_{LO} + 55x_{LF} + 69x_{LS} + 49x_{NF} + 78x_{FS} + 47x_{BN} + 30x_{BF} + 100x_{BS}.$$

For each node, the flow conservation constraints are

$$\begin{aligned} x_{ON} - x_{LO} &= s_o = -125, & (\text{demand node}) \\ x_{LO} + x_{LF} + x_{LS} &= s_L = 350 - 125 = 225, & (\text{supply node}) \\ x_{NF} - x_{ON} - x_{BN} &= s_N = -125, & (\text{demand node}) \\ x_{FS} - x_{NF} - x_{BF} - x_{LF} &= s_F = -125, & (\text{demand node}) \\ x_{BN} + x_{BS} + x_{BF} &= s_B = 400 - 125 = 275, & (\text{supply node}) \\ -x_{BS} - x_{FS} - x_{LS} &= s_S = -125. & (\text{demand node}) \end{aligned}$$

Note that, for the two cities where the units are produced, we have deduced the local demand that does not need to be transported:

- Lausanne: $350 - 125 = 225$,
- Bern: $400 - 125 = 275$.

The capacity constraints on each arc (i, j) are

$$0 \leq x_{ij} \leq 200.$$

Putting everything together, we have

$$\min_{x \in \mathbb{R}^9} 34x_{ON} + 27x_{LO} + 55x_{LF} + 69x_{LS} + 49x_{NF} + 78x_{FS} + 47x_{BN} + 30x_{BF} + 100x_{BS},$$

subject to

$$\begin{aligned} x_{ON} - x_{LO} &= -125, \\ x_{LO} + x_{LF} + x_{LS} &= 225 \\ x_{NF} - x_{ON} - x_{BN} &= -125 \\ x_{FS} - x_{NF} - x_{BF} - x_{LF} &= -125 \\ x_{BN} + x_{BS} + x_{BF} &= 275 \\ -x_{BS} - x_{FS} - x_{LS} &= -125 \end{aligned}$$

$$\begin{array}{ll}
x_{ON} \geq 0, & x_{ON} \leq 200, \\
x_{LO} \geq 0, & x_{LO} \leq 200, \\
x_{LF} \geq 0, & x_{LF} \leq 200, \\
x_{LS} \geq 0, & x_{LS} \leq 200, \\
x_{NF} \geq 0, & x_{NF} \leq 200, \\
x_{BN} \geq 0, & x_{BN} \leq 200, \\
x_{BF} \geq 0, & x_{BF} \leq 200, \\
x_{BS} \geq 0, & x_{BS} \leq 200, \\
x_{FS} \geq 0, & x_{FS} \leq 200.
\end{array}$$

Note that the sum of elements of the vector s is equal to 0:

$$-125 + 225 - 125 - 125 + 275 - 125 = 0.$$

2. The standard form is

$$\min c^T x$$

subject to

$$\begin{array}{l}
Ax = b, \\
x \geq 0.
\end{array}$$

The objective function is already in standard form (i.e. minimization). We need to modify the constraints. We need to include a slack variable for each inequality constraint different from $x_{ij} \geq 0$. We obtain the following problem in standard form:

$$\min 34x_{ON} + 27x_{LO} + 55x_{LF} + 69x_{LS} + 49x_{NF} + 78x_{FS} + 47x_{BN} + 30x_{BF} + 100x_{BS},$$

subject to

$$\begin{aligned}
x_{ON} - x_{LO} &= -125, \\
x_{LO} + x_{LF} + x_{LS} &= 225, \\
x_{NF} - x_{ON} - x_{BN} &= -125, \\
x_{FS} - x_{NF} - x_{BF} - x_{LF} &= -125, \\
x_{BN} + x_{BS} + x_{BF} &= 275, \\
-x_{BS} - x_{FS} - x_{LS} &= -125, \\
x_{ON} + y_{ON} &= 200, \\
x_{LO} + y_{LO} &= 200, \\
x_{LF} + y_{LF} &= 200, \\
x_{LS} + y_{LS} &= 200, \\
x_{NF} + y_{NF} &= 200, \\
x_{BN} + y_{BN} &= 200, \\
x_{BF} + y_{BF} &= 200, \\
x_{BS} + y_{BS} &= 200, \\
x_{FS} + y_{FS} &= 200,
\end{aligned}$$

$$\begin{aligned}
x_{ON} &\geq 0, & y_{ON} &\geq 0, \\
x_{LO} &\geq 0, & y_{LO} &\geq 0, \\
x_{LF} &\geq 0, & y_{LF} &\geq 0, \\
x_{LS} &\geq 0, & y_{LS} &\geq 0, \\
x_{NF} &\geq 0, & y_{NF} &\geq 0, \\
x_{BN} &\geq 0, & y_{BN} &\geq 0, \\
x_{BF} &\geq 0, & y_{BF} &\geq 0, \\
x_{BS} &\geq 0, & y_{BS} &\geq 0, \\
x_{FS} &\geq 0, & y_{FS} &\geq 0.
\end{aligned}$$

If we number the arcs using the following order: $x_{ON}, x_{LO}, x_{LF}, x_{LS}, x_{NF}, x_{FS}, x_{BN}, x_{BF}, x_{BS}, y_{ON}, y_{LO}, y_{LF}, y_{LS}, y_{NF}, y_{FS}, y_{BN}, y_{BF}$,

y_{BS} , the non zero entries of the matrix A are

$$A = \left(\begin{array}{cccccccc|cccccccc} 1 & -1 & & & & & & & & & & & & & & & \\ & & 1 & 1 & 1 & & & & & & & & & & & & \\ -1 & & & & & 1 & & -1 & & & & & & & & & \\ & & & -1 & & -1 & 1 & & -1 & & & & & & & & \\ & & & & & & & 1 & 1 & 1 & & & & & & & \\ & & & & -1 & & -1 & & & & -1 & & & & & & \\ \hline & 1 & & & & & & & & & 1 & & & & & & \\ & & 1 & & & & & & & & & 1 & & & & & \\ & & & 1 & & & & & & & & & 1 & & & & \\ & & & & 1 & & & & & & & & & 1 & & & \\ & & & & & 1 & & & & & & & & & 1 & & \\ & & & & & & 1 & & & & & & & & & 1 & \\ & & & & & & & 1 & & & & & & & & & 1 \\ & & & & & & & & 1 & & & & & & & & \\ & & & & & & & & & 1 & & & & & & & \\ & & & & & & & & & & 1 & & & & & & \\ & & & & & & & & & & & 1 & & & & & \\ & & & & & & & & & & & & 1 & & & & \\ & & & & & & & & & & & & & 1 & & & \\ & & & & & & & & & & & & & & 1 & & \\ & & & & & & & & & & & & & & & 1 & \\ & & & & & & & & & & & & & & & & 1 \end{array} \right).$$

As we will see later in this chapter, the structure of the matrix A plays an important role in the properties of the transshipment problem. In particular, we will exploit the property that each column contains exactly two non zero entries: 1 and -1. The matrix presented above does not have this property. We will therefore apply a simple substitution based on the constraints $x_{ij} + y_{ij} = 200$. Each x_{ij} appearing with a positive sign in the first set of constraints is replaced by $200 - y_{ij}$. We obtain an equivalent standard form formulation, with the following set

of constraints:

$$\begin{aligned}
-y_{ON} - x_{LO} &= -125 - 200 = -325, \\
-y_{LO} - y_{LF} - y_{LS} &= 225 - 600 = -375, \\
-y_{NF} - x_{ON} - x_{BN} &= -125 - 200 = -325, \\
-y_{FS} - x_{NF} - x_{BF} - x_{LF} &= -125 - 200 = -325, \\
-y_{BN} - y_{BS} - y_{BF} &= 275 - 600 = -325, \\
-x_{BS} - x_{FS} - x_{LS} &= -125, \\
x_{ON} + y_{ON} &= 200, \\
x_{LO} + y_{LO} &= 200, \\
x_{LF} + y_{LF} &= 200, \\
x_{LS} + y_{LS} &= 200, \\
x_{NF} + y_{NF} &= 200, \\
x_{BN} + y_{BN} &= 200, \\
x_{BF} + y_{BF} &= 200, \\
x_{BS} + y_{BS} &= 200, \\
x_{FS} + y_{FS} &= 200,
\end{aligned}$$

$$\begin{aligned}
x_{ON} &\geq 0, & y_{ON} &\geq 0, \\
x_{LO} &\geq 0, & y_{LO} &\geq 0, \\
x_{LF} &\geq 0, & y_{LF} &\geq 0, \\
x_{LS} &\geq 0, & y_{LS} &\geq 0, \\
x_{NF} &\geq 0, & y_{NF} &\geq 0, \\
x_{BN} &\geq 0, & y_{BN} &\geq 0, \\
x_{BF} &\geq 0, & y_{BF} &\geq 0, \\
x_{BS} &\geq 0, & y_{BS} &\geq 0, \\
x_{FS} &\geq 0, & y_{FS} &\geq 0.
\end{aligned}$$

The matrix A has now the property that each column contains exactly two

non zero entries, a 1 and a -1.

$$A = \left(\begin{array}{cccccc|cccccc} & & & & & & -1 & & & & & \\ & & & & & & & -1 & -1 & -1 & & \\ & -1 & & & & & & & & & -1 & \\ & & & & & & & & & & & -1 \\ & & & -1 & & -1 & & & & & & \\ & & & & & & & & & & & -1 \\ & & & & -1 & & -1 & & & & & \\ \hline & 1 & & & & & & & & & & \\ & & 1 & & & & & & & & & \\ & & & 1 & & & & & & & & \\ & & & & 1 & & & & & & & \\ & & & & & 1 & & & & & & \\ & & & & & & 1 & & & & & \\ & & & & & & & 1 & & & & \\ & & & & & & & & 1 & & & \\ & & & & & & & & & 1 & & \\ & & & & & & & & & & 1 & \\ & & & & & & & & & & & 1 \end{array} \right).$$

The vector b is

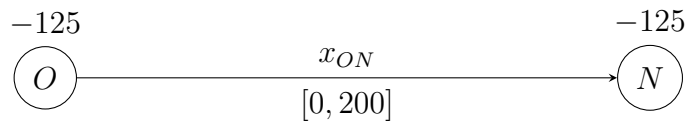
$$b = \begin{pmatrix} -325 \\ -375 \\ -325 \\ -325 \\ -325 \\ -125 \\ 200 \\ 200 \\ 200 \\ 200 \\ 200 \\ 200 \\ 200 \\ 200 \\ 200 \end{pmatrix}.$$

Note that it can be interpreted as a set of supply and demand, as its entries

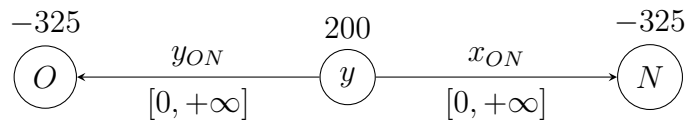
sum up to zero. The vector c is

$$c = \begin{pmatrix} 34 \\ 27 \\ 55 \\ 69 \\ 49 \\ 78 \\ 47 \\ 30 \\ 100 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Interestingly, this formulation can be seen as a standard transshipment problem on a modified network. To see this, consider the arc ON . In the original formulation, the flow on that arc was constrained between 0 and 200. Both O and N are demand nodes, with $s_O = s_N = -125$.



Each slack variable can be associated with a new node and a new arc, in the following configuration.



The interpretation is that we inject 200 units of flow in the network at the node corresponding to the slack variable. By construction, the flow x_{ON}

cannot exceed 200, and there is no need to set an upper bound on that arc. The new arc, associated with the slack variable, carries the remaining flow. The constraints $x_{ij} + y_{ij} = 200$ are the flow conservation constraints for the new nodes.

The optimal solution of the problem in standard form is

$$\begin{array}{ll}
 x_{ON} = 75, & y_{ON} = 125, \\
 x_{LO} = 200, & y_{LO} = 0, \\
 x_{LF} = 0, & y_{LF} = 200, \\
 x_{LS} = 25, & y_{LS} = 175, \\
 x_{NF} = 0, & y_{NF} = 200, \\
 x_{FS} = 0, & y_{FS} = 200, \\
 x_{BN} = 50, & y_{BN} = 150, \\
 x_{BF} = 125, & y_{BF} = 75, \\
 x_{BS} = 100, & y_{BS} = 100,
 \end{array}$$

and is represented on the following picture, where the number associated with each arc is the flow:

